

AI-Based Traffic Sign Detection, Localization and Map Dashboard System

Project Report (Final)

*Submission-ready report. Citation labels [n] refer to
REFERENCES_FINAL.md.*

*All technical claims reflect the actual repository (migrations 0001–0006,
the*

*src/ codebase, and the Playwright E2E suite). The detection model is
external and is*

*not trained or benchmarked in this project; no detection-accuracy or
benchmark numbers*

*are claimed. Demo counts cited in §17 are representative seeded-dataset
counts, not*

performance measurements.

Abstract

This project presents a web platform that converts raw, per-frame traffic-sign detections produced by an external computer-vision model into a structured, location-aware, deduplicated, and reviewable traffic-sign inventory visualised on an interactive map. Field devices capture camera frames together with GPS metadata; an external (or, for development and demonstration, a deterministic mock) detector returns sign detections; and the backend persists every detection, consolidates repeated observations of the same physical sign through a distance- and confidence-weighted localisation method, and exposes maps, analytics, review workflows, and operational tooling to administrators. The detection model is deliberately treated as an external inference service; the contribution of this work is the surrounding platform — integration, geolocation fusion, duplicate filtering, secure media delivery,

observability, and data-lifecycle management. The system is implemented with Next.js 14 (App Router) [10], TypeScript, Tailwind CSS, and Supabase (Authentication, PostgreSQL, and Storage) [11]. It was validated against a real Supabase backend with migrations 0001–0006 applied, a private storage bucket, an administrator account, successful demo-data seeding, an end-to-end (E2E) test suite passing 24 of 24 cases, and a clean lint, type-check, and production build.

1. Introduction

Road safety remains a major global public-health concern [14], and accurate road-asset information — including an up-to-date inventory of traffic signs — supports infrastructure maintenance and planning. Building such an inventory manually is slow and error-prone. Automatic traffic-sign detection and recognition from imagery have therefore been studied extensively [1], [2], and modern object detectors — for example YOLO [3], Faster R-CNN [4], and SSD [5] — can localise and classify signs in images. However, a raw stream of per-frame detections is not itself an inventory: the same physical sign is detected repeatedly, GPS readings vary, and confidence fluctuates. This project addresses the gap between *detection* and a *usable inventory* by building an end-to-end web system that ingests detections, fuses them with geolocation, removes duplicates, and presents a verified, queryable map.

2. Problem Definition

Given a stream of camera frames with GPS metadata and an external detector that returns, per detection, a class, a confidence value, and a bounding box, the system must produce and maintain a single, deduplicated, location-refined, and reviewable record for each physical sign, while preserving the complete raw detection history for audit. Secondary requirements include authentication and authorisation, private media delivery, operational observability, and reproducible demonstration.

3. Project Motivation

The project aims to (i) transform a noisy detection stream into decision-ready data; (ii) demonstrate a realistic, production-shaped architecture — external model service, row-level security, signed media, observability, and scheduled maintenance — rather than a minimal prototype; and (iii) provide a defensible, full-stack academic artefact spanning web engineering, data modelling, geospatial processing, and operational governance.

4. System Requirements

Functional requirements: authentication and roles; device registration; camera/GPS detection sessions; AI integration through a defined contract; persistence of raw detection events; duplicate grouping and location refinement; sign and device maps; administrative review, logs, user and device management; AI health, self-test, and observability; analytics with durable daily snapshots; storage governance; demo seeding; and a presentation mode.

Non-functional requirements: security (authentication, row-level security, private storage with signed URLs [11], [12], and server-only secrets); reliability (request timeout, bounded retry, and response validation in the AI layer); observability (an audit log and administrative dashboards); maintainability (typed code, ordered migrations, and automated smoke tests); and demonstrability.

Actors: the field user, the administrator, the external AI model service, and the scheduler/cron caller.

5. Related Work

Automatic traffic-sign recognition and detection are established research areas, with public benchmarks such as the German Traffic Sign Recognition Benchmark [1] and the German Traffic Sign Detection Benchmark [2], and with general-purpose detectors such as YOLO [3], Faster R-CNN [4], and SSD [5] frequently applied to the task. The duplicate-

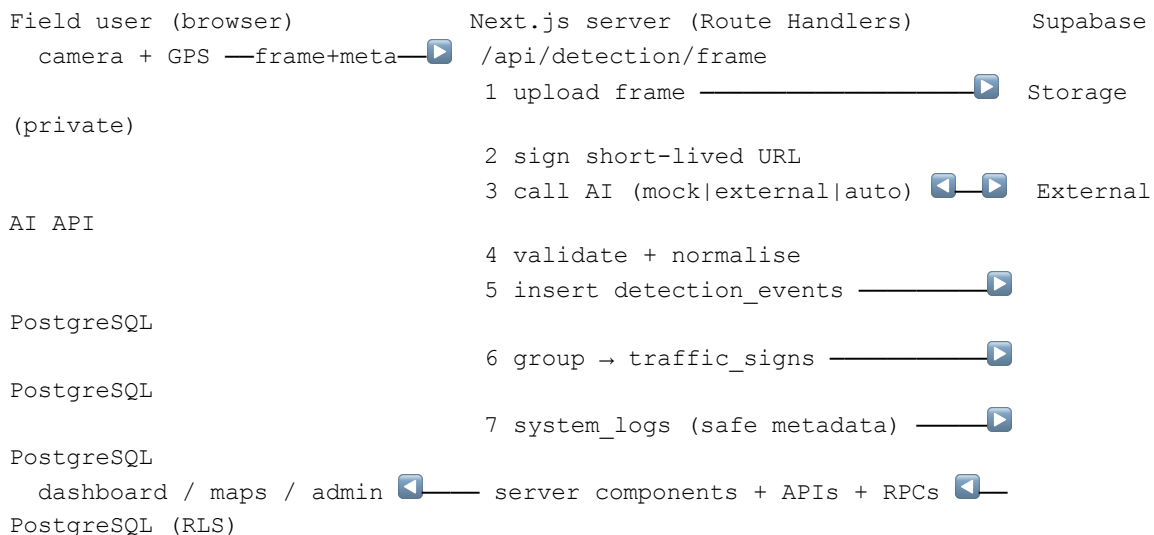
consolidation problem addressed here is conceptually related to density-based spatial clustering [7], which groups nearby points and isolates noise. Human-in-the-loop review of machine-learning outputs — reflected in this project's administrative verification workflow — is a recognised approach for improving the operational quality of model outputs [16]. A detailed comparative survey is left as future work.

6. System Architecture

The application uses the Next.js App Router [10] with a `(protected)` route group whose server layout re-checks the session (a defence-in-depth measure behind middleware) and renders a role-aware navigation sidebar. Server Components and Route Handlers execute on the server, while client components handle interactivity (camera, maps, tables, forms) and communicate with Route Handlers over HTTP. The server operates in two trust tiers:

- a **row-level-security (RLS) client** bound to the user's session for user-scoped reads [12]; and

- a **service-role client**, used only on the server for trusted writes and administrative/analytics database functions, and always after an explicit authentication and role check.



7. Technology Stack

The platform is built with Next.js 14 (App Router) [10], TypeScript, and Tailwind CSS on

the frontend; Supabase (Authentication, PostgreSQL, Storage, and row-level security) [11], [12] on the backend; Leaflet with OpenStreetMap tiles for mapping [8], [9]; and Playwright for end-to-end testing. Charts are rendered with lightweight Tailwind/SVG components rather than an external charting library, to minimise dependencies.

8. Data Model and Database Design

The schema is defined across six ordered SQL migrations:

Migration	Purpose
0001_init.sql	Core tables, RLS policies, a profile-creation trigger, and an admin helper.
0002_secure_image_paths.sql	Storage object-path columns (image_path, representative_image_path).
0003_backfill_image_paths.sql	Idempotent backfill of legacy public-URL columns into path columns.
0004_analytics_rpc.sql	Service-role analytics functions (AI summary, breakdown, time-series, dashboard summary).
0005_storage_quarantine.sql	storage_quarantine_candidates and storage_reconciliation_runs.
0006_daily_metrics_snapshots.sql	daily_metrics_snapshots and a snapshot upsert function.

Core tables include profiles, devices, detection_sessions, detection_events, traffic_signs, traffic_sign_observations, device_location_logs, and system_logs, alongside the governance and analytics tables introduced above. The design uses UUID primary keys, timestampz timestamps, and jsonb columns for raw AI responses and metadata. Row-level security is enabled on all tables; the maintenance and analytics tables expose no client policies and are reached only through the service-role client [12].

9. AI Model API Integration

The AI layer defines a canonical request/response contract and supports three modes, selected by the AI_MODEL_MODE environment variable:

- ``mock`` — a built-in deterministic detector that performs no network call (used for development, demonstrations, and tests);

- ``external`` — always calls the configured model endpoint, returning a clear error if no endpoint is configured; and
- ``auto`` — calls the endpoint when configured and otherwise falls back to the mock detector, without silently falling back after a *failed* external call.

A request carries a short-lived signed image URL together with session, device, timestamp, and location metadata, and contains no secrets. Responses are validated and normalised — the detections field must be an array, each confidence value must lie in $[0, 1]$, each class name must be non-empty, and bounding boxes must be well-formed — before any database write, so that an invalid response cannot corrupt or crash the pipeline. External calls are bounded by a request timeout and retried only on transient failures (timeout, network error, or HTTP 429/502/503/504) with linear backoff. General-purpose detectors such as YOLO [3], Faster R-CNN [4], and SSD [5] are representative of the kind of model that such an external endpoint might host; the platform itself is model-agnostic.

10. Detection Session Workflow

A detection session proceeds as follows: (1) the user selects a registered, active device and starts a session; (2) the browser captures a downscaled JPEG frame and the current GPS reading and posts them to the frame endpoint; (3) the server uploads the frame to the private bucket and stores its object path; (4) a short-lived signed URL is sent to the AI layer for inference; (5) the normalised response is persisted as one detection-event row per detection, with the raw JSON retained, a device location-log entry written, and session counters updated; (6) grouping and localisation consolidate the detection into the sign inventory; and (7) safe audit entries are written to the system log.

11. Geolocation and Localisation Method

Each detection carries latitude, longitude, GPS accuracy, heading, and speed. The

representative location of a sign is refined from all of its observations using a

confidence- and accuracy-weighted average:

```
weight = confidence / max(gps_accuracy, 1)
refined_location =  $\Sigma(\text{weight} \cdot \text{position}) / \Sigma(\text{weight})$ 
```

This formulation reduces the influence of low-confidence detections and of imprecise GPS fixes when computing a sign's coordinates. Satellite positioning has a bounded but non-zero error budget — the GPS Standard Positioning Service commits to a global average user range error within a specified limit rather than to exact positions [15] — so a single GPS fix should not be treated as the exact location of a sign; weighting by reported accuracy and averaging across observations mitigates this uncertainty.

12. Duplicate Detection and Traffic Sign Grouping

For each saved detection whose confidence exceeds a configured minimum, the system:

1. retrieves existing signs of the **same sign type**;
2. computes the **Haversine** (great-circle) distance to each candidate [6] and, if the nearest candidate lies within `SIGN_MATCH_RADIUS_METERS` (default 25 m), records the detection as an additional observation of that sign; otherwise it creates a new sign;
3. recomputes the sign's location using the weighted average of §11; and
4. **auto-verifies** a sign once it has at least three observations and an average confidence above 0.75, without overriding any manual decision.

This same-type, proximity-bounded consolidation is conceptually aligned with density-based spatial clustering [7], specialised here to the traffic-sign domain. Verification states include `pending`, `auto_verified`, `manually_verified`, `rejected`, `duplicate`, and `low_confidence`.

13. Map Visualisation and Dashboard

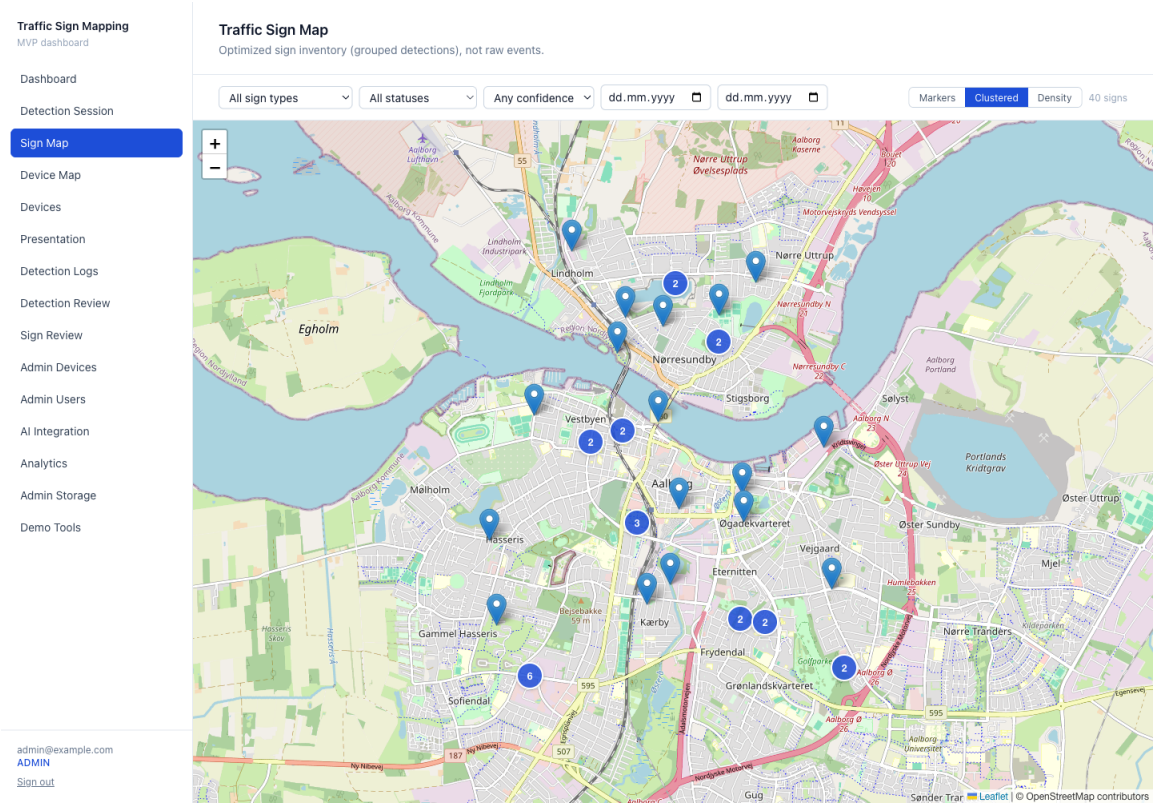


Figure: Static traffic-sign inventory map (clustered markers).

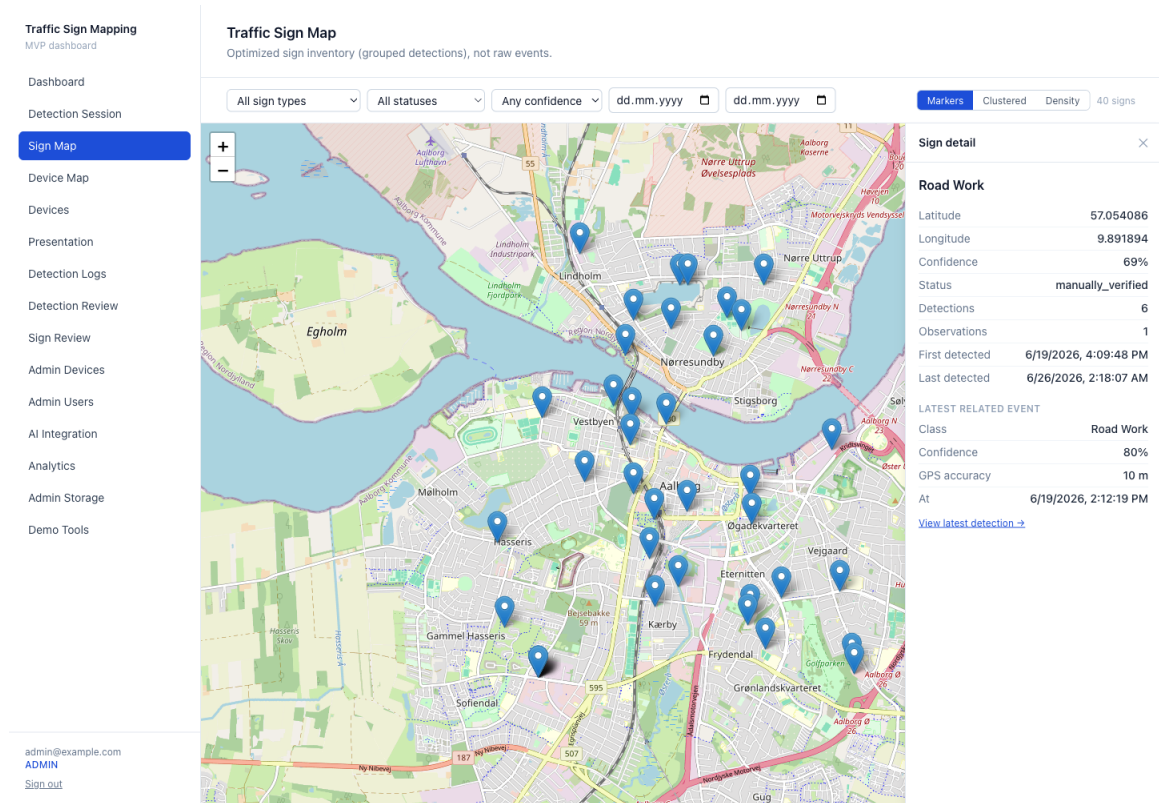


Figure: Sign detail panel for a selected marker.

The static sign map renders the optimised inventory with markers, zoom-aware clustering, and a density view, together with filters (type, status, confidence, and date) and a detail panel; clustering and density are computed within the application and require no additional mapping dependency. A separate live device map shows last-known device positions. The dashboard presents key performance indicators sourced from a database-side function (with a JavaScript fallback), a verification breakdown, and the most frequent sign types. Maps use Leaflet [8] with OpenStreetMap tiles [9] and are imported dynamically on the client to avoid server-rendering issues.

14. Admin Management and Observability

Traffic Sign Mapping

MVP dashboard

Dashboard

Detection Session

Sign Map

Device Map

Devices

Presentation

Detection Logs

Detection Review

Sign Review

Admin Devices

Admin Users

AI Integration

Analytics

Admin Storage

Demo Tools

Admin - Detections

Review raw AI detection events: verify, reject, mark duplicate, or reset.

Export events CSVExport signs CSV

Search class name...

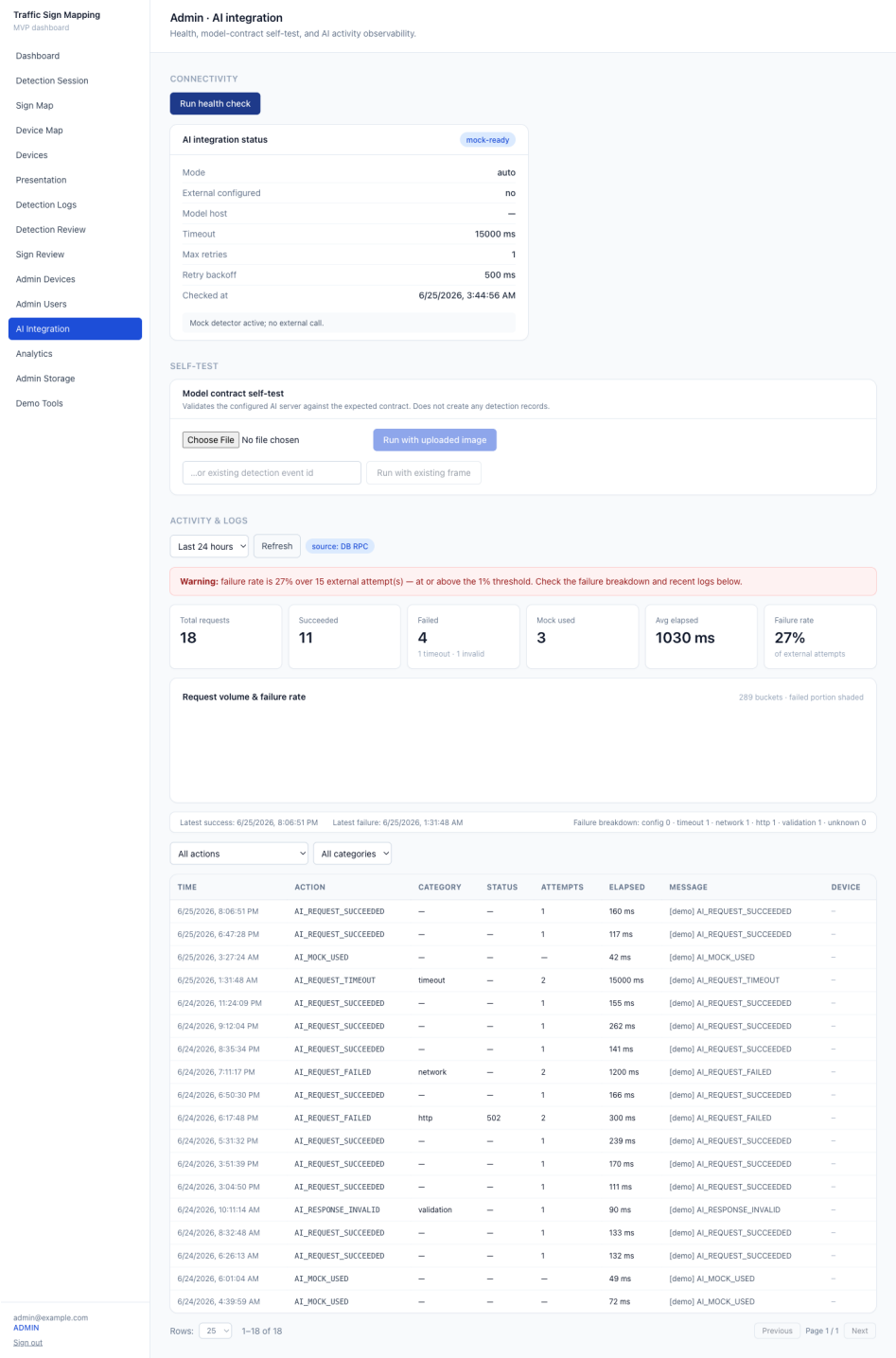
All statuses

FRAME	CLASS	CONF.	DEVICE	USER	LAT	LNG	ACC.	AI MS	IMG	STATUS	TIME	
—	Speed Limit 50	61%	Dashcam – Van 7	Demo Admin	57.04537	9.91127	12m	85	—	pending	6/26/2026, 2:09:30 AM	View details Verify Reject Duplicate
—	Stop	67%	IoT Node – Bridge	Demo Admin	57.02995	9.93573	7m	118	—	pending	6/26/2026, 1:47:19 AM	View details Verify Reject Duplicate
—	No Entry	57%	Vehicle Cam – Patrol 1	Demo Admin	57.05849	9.95410	6m	182	—	pending	6/25/2026, 11:04:24 PM	View details Verify Reject Duplicate
—	School Zone	91%	IoT Node – Bridge	Demo Admin	57.06431	9.91961	6m	202	—	pending	6/25/2026, 9:53:23 PM	View details Verify Reject Duplicate
—	Roundabout	65%	Dashcam – Van 7	Demo Admin	57.06123	9.90961	12m	153	—	pending	6/25/2026, 8:31:27 PM	View details Verify Reject Duplicate
—	Speed Limit 50	88%	Field Phone – Surveyor	Demo Admin	57.06585	9.93160	10m	217	—	pending	6/25/2026, 6:29:25 PM	View details Verify Reject Duplicate
—	Parking	70%	Dashcam – Van 7	Demo Admin	57.06125	9.92860	6m	152	—	pending	6/25/2026, 4:07:04 PM	View details Verify Reject Duplicate
—	Stop	70%	Vehicle Cam – Patrol 1	Demo Admin	57.02544	9.95822	11m	159	—	rejected	6/25/2026, 3:35:11 PM	View details Verify Reject Duplicate
—	Roundabout	87%	Vehicle Cam – Patrol 1	Demo Admin	57.07303	9.89971	5m	187	—	pending	6/25/2026, 1:56:46 PM	View details Verify Reject Duplicate
—	Stop	68%	Dashcam – Van 7	Demo Admin	57.02548	9.95803	4m	173	—	manually_verified	6/25/2026, 1:06:48 PM	View details Verify Reject Duplicate
—	No Entry	90%	Vehicle Cam – Patrol 1	Demo Admin	57.03383	9.95585	8m	214	—	pending	6/25/2026, 9:45:36 AM	View details Verify Reject Duplicate
—	Pedestrian Crossing	68%	IoT Node – Bridge	Demo Admin	57.04320	9.92306	4m	225	—	pending	6/25/2026, 8:29:01 AM	View details Verify Reject Duplicate
—	Speed Limit 30	49%	IoT Node – Bridge	Demo Admin	57.06939	9.93933	5m	230	—	pending	6/25/2026, 7:18:39 AM	View details Verify Reject Duplicate
—	Parking	86%	IoT Node – Bridge	Demo Admin	57.03768	9.91494	4m	111	—	pending	6/25/2026, 7:11:42 AM	View details Verify Reject Duplicate
—	Stop	72%	Vehicle Cam – Patrol 1	Demo Admin	57.03309	9.94304	11m	282	—	pending	6/25/2026, 6:45:47 AM	View details Verify Reject Duplicate
—	Speed Limit 50	58%	Vehicle Cam – Patrol 1	Demo Admin	57.03440	9.92102	9m	236	—	pending	6/25/2026, 4:51:48 AM	View details Verify Reject Duplicate
—	Parking	59%	Field Phone – Surveyor	Demo Admin	57.03743	9.91470	11m	94	—	pending	6/25/2026, 4:12:36 AM	View details Verify Reject Duplicate
—	Roundabout	51%	Dashcam – Van 7	Demo Admin	57.06140	9.90948	11m	265	—	pending	6/25/2026, 2:01:25 AM	View details Verify Reject Duplicate
—	Speed Limit 30	81%	IoT Node – Bridge	Demo Admin	57.06969	9.93950	3m	104	—	pending	6/25/2026, 1:17:36 AM	View details Verify Reject Duplicate
—	Speed Limit 50	82%	Vehicle Cam – Patrol 1	Demo Admin	57.06945	9.92290	13m	71	—	pending	6/25/2026, 12:24:19 AM	View details Verify Reject Duplicate
—	Stop	77%	IoT Node – Bridge	Demo Admin	57.02514	9.95803	6m	207	—	rejected	6/24/2026, 11:57:50 PM	View details Verify Reject Duplicate
—	Speed Limit 50	85%	Vehicle Cam – Patrol 1	Demo Admin	57.04535	9.91140	14m	246	—	manually_verified	6/24/2026, 10:48:36 PM	View details Verify Reject Duplicate
—	Roundabout	76%	Vehicle Cam – Patrol 1	Demo Admin	57.04675	9.90103	7m	294	—	rejected	6/24/2026, 8:45:20 PM	View details Verify Reject Duplicate
—	No Entry	76%	Field Phone – Surveyor	Demo Admin	57.05336	9.91851	10m	192	—	pending	6/24/2026, 5:11:12 PM	View details Verify Reject Duplicate
—	Road Work	41%	IoT Node – Bridge	Demo Admin	57.05410	9.91118	13m	205	—	low_confidence	6/24/2026, 4:01:13 PM	View details Verify Reject Duplicate

Rows: 251–25 of 120

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Figure: Admin per-detection review table.



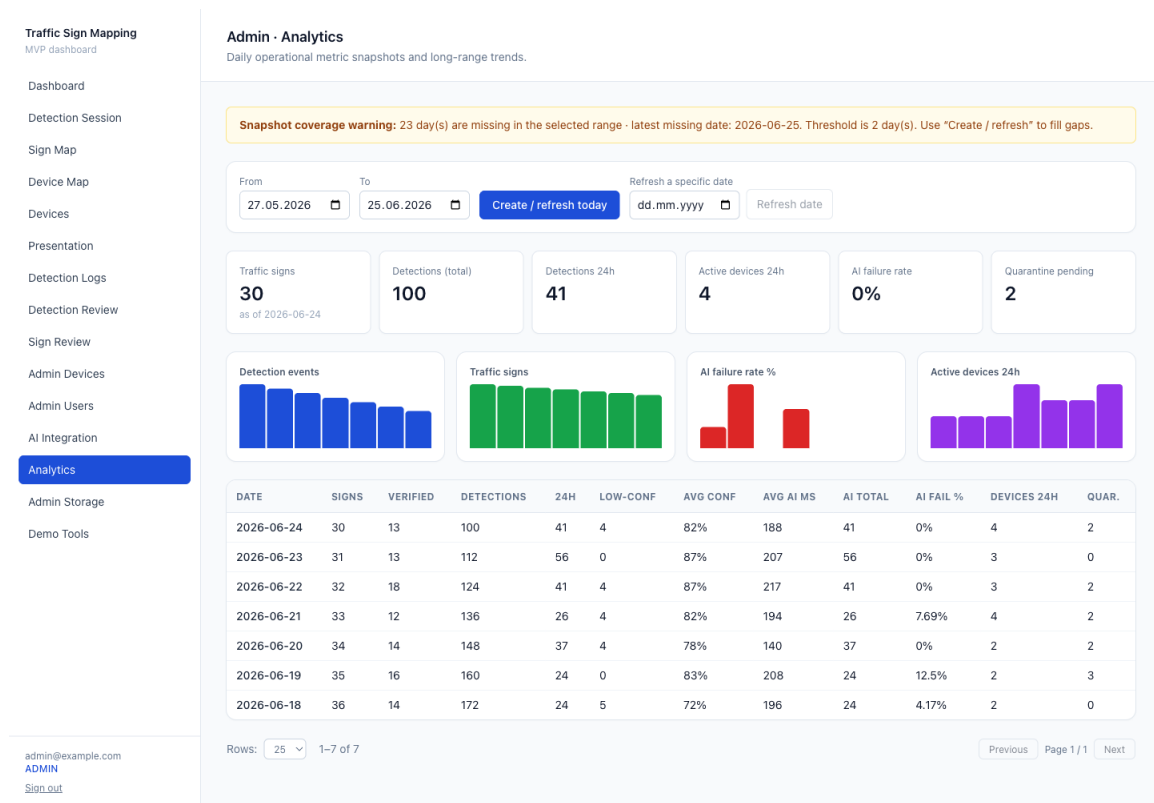


Figure: Daily metric snapshots and trends.

Administrators manage users — including Supabase Authentication user creation and password reset — devices, and the review of detections and signs. The AI console provides a connectivity health check, a model-contract self-test that creates no production data, and observability views (an activity summary, a failure breakdown, a time-series, and a failure-rate threshold alert) backed by database functions with a JavaScript fallback. The analytics view presents durable daily metric snapshots, trend bars, and snapshot-coverage gap detection. This administrative verification capability implements human-in-the-loop review intended to improve the operational quality of the model's outputs [16]; it supports manual correction and curation but does not, by itself, guarantee correctness. All administrative APIs return HTTP 401 for unauthenticated callers and HTTP 403 for authenticated non-administrators.

15. Storage Security and Signed-URL Design

Traffic Sign Mapping

MVP dashboard

Dashboard

Detection Session

Sign Map

Device Map

Devices

Presentation

Detection Logs

Detection Review

Sign Review

Admin Devices

Admin Users

AI Integration

Analytics

Admin Storage

Demo Tools

Admin - Storage

Backfill status, quarantine-first reconciliation, and safe orphan cleanup.

BACKFILL & LEGACY TOOLS

Image path backfill status

Events total	Events w/ path	Events legacy-only	Signs total	Signs w/ path	Signs legacy-only
120	0	0	40	0	0

Backfill actions

Dry run

Apply backfill

Orphaned storage scan

Scan orphans

Delete selected (0)

Storage security notes

- The bucket should be **private** in production.
- Images are served via short-lived signed URLs minted server-side.
- Expired image URLs can be refreshed in-app (no full page reload needed).
- CSV exports never include signed URLs; logs never store signed URLs.
- Cleanup only touches unreferenced objects under sessions/.

QUARANTINE RECONCILIATION (RECOMMENDED)

Reconciliation

Grace period: 7 day(s) - scan caps: 100 folders × 200 files - delete batch ≤ 50 - prefix sessions/

Reconciliation only records unreferenced objects as pending quarantine candidates. It never deletes anything.

Run reconciliation scan

Recent runs

No reconciliation runs recorded yet.

Quarantine candidates

Search path...

pending

☐ Eligible only

Delete selected (0)

No quarantine candidates.

Rows: 25

0-0 of 0

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Figure: Storage governance — quarantine-first reconciliation.

The storage bucket is private. The database stores the storage object path rather than a public URL, and every API that returns an image mints a short-lived signed URL on the server, only after authorisation (owner-or-administrator for detection events; RLS-readable for shared signs) [11]. A dedicated endpoint re-signs images by entity identifier — never from a client-supplied path — so that expired images can be refreshed in place. CSV exports emit an availability flag and the object path, never a signed URL. Governance tooling adds legacy-path backfill, quarantine-first reconciliation (unreferenced objects become pending candidates; deletion is grace-period gated,

re-checked immediately before removal, and administrator-initiated — never automatic), and a record of reconciliation runs. Secret-protected scheduled endpoints (authenticated by a bearer secret) enable daily snapshots and reconciliation scans without deleting objects.

16. Demo Data and Testing Strategy

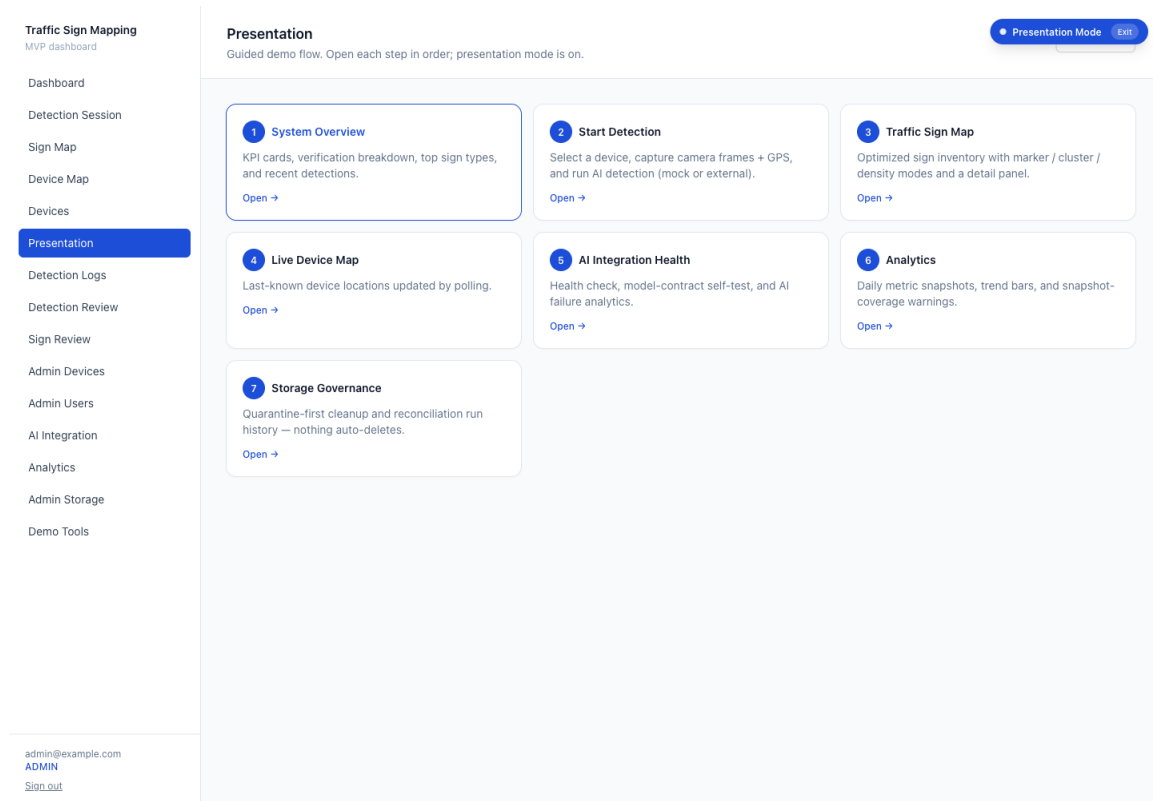


Figure: Guided presentation mode.

A deterministic, demo-marked seeder creates devices, sessions, detections, signs, observations, location logs, AI logs, and snapshots; the clear operation removes only demo-marked data. A presentation mode and a guided presentation page support demonstrations without bypassing authentication. Automated testing uses Playwright, organised into an unauthenticated suite (route and API gating, and cron-secret handling), an authenticated suite (the dashboard, all administrative pages, both maps, the presentation page, and the detection page with camera and GPS mocked), and an opt-in demo-seed suite. The tests are environment-aware and skip cleanly when

credentials are absent.

17. Evaluation and Validation

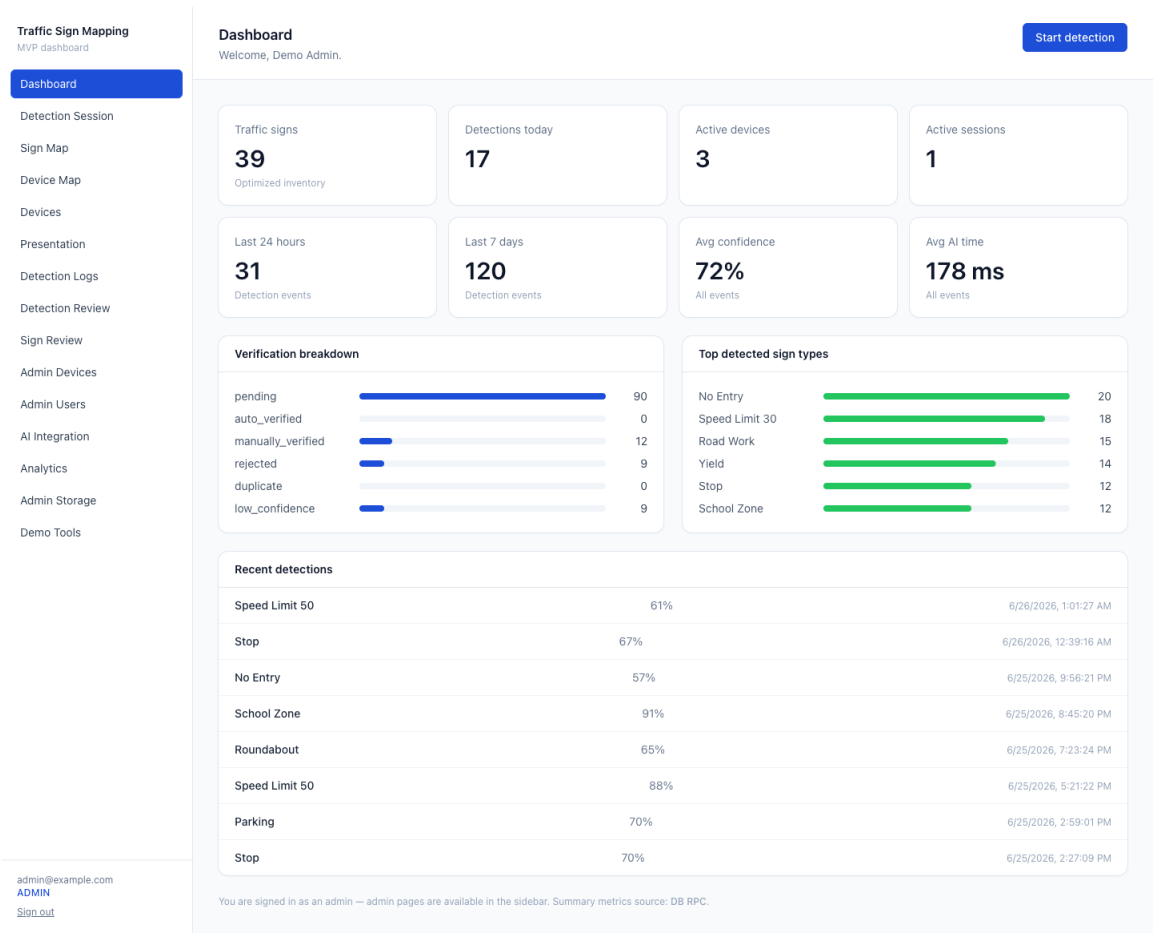


Figure: Dashboard KPIs sourced from a DB-side RPC.

The system was validated against a real Supabase backend, with the following verified outcomes:

- migrations 0001–0006 applied, and all expected tables, columns, and the analytics function confirmed present;
- a private storage bucket configured (public access disabled);
- an administrator account present with the administrator role;
- demo data seeded successfully (in a representative run: 4 devices, 6 sessions, 120 detection events, 35 signs, and 7 daily snapshots — these are representative seeded dataset counts, not performance measurements);

- a Playwright end-to-end suite passing 24 of 24 cases against the Supabase-backed environment; and
- a clean lint, type-check, and production build (54 application routes built).

The accuracy of the detection model is not evaluated in this work because the model is an external service and no labelled evaluation dataset was measured here; benchmark datasets such as [1] and [2] would support such an evaluation in future work.

18. Limitations

- Sign matching performs a same-type scan without a spatial index; very large inventories would benefit from spatial indexing (for example, PostGIS [13]).
- Maps load a bounded number of signs on the client; larger datasets would require server-side tiling or aggregation.
- Some analytics use a bounded in-application fallback when the database functions are unavailable.
- Demo detections use null image paths (no fabricated signed URLs), so their detail views display a graceful "no image" state.
- No detection-model accuracy evaluation is performed, as the model is external.
- Reconciliation scans are prefix- and count-bounded and are therefore not exhaustive for very large buckets.

19. Future Work

- Introduce spatial indexing (for example, PostGIS [13]) and server-side map tiling for scalability.
- Conduct a labelled evaluation of grouping and localisation quality using public benchmarks such as [2].
- Add a background reconciliation reaper and alerting, and rotate the cron secret.
- Expand automated test coverage and run it in continuous integration against a dedicated test project.

- Support model versioning and multiple model backends.

20. Conclusion

The project delivers a complete, validated pipeline that transforms noisy per-frame detections into a deduplicated, location-refined, reviewable, and securely served traffic-sign inventory, complemented by administrative observability and reproducible demonstration. By treating the detector as an external service, the work concentrates on the integration, geospatial, security, and operational problems that make detections operationally useful.

21. References

The full bibliography ([1]–[16]) is provided in `REFERENCES_FINAL.md`. Confirm all entries against the citation style required by your institution before final submission.

References (Final)

Final bibliography for PROJECT_REPORT_FINAL.md. Academic entries ([1]–[7], [16]) and official sources ([14], [15]) were checked against indexing services / publisher / official pages during preparation; software entries ([8]–[13]) are official project documentation. No DOIs, URLs, authors, titles, venues, or years were fabricated; where a DOI was not confirmed it is omitted. Web references use an "accessed 2026" retrieval note; reformat to the exact required citation style (IEEE/APA/Harvard) before submission.

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DOI: 10.1016/j.future.2022.05.014 [Peer-reviewed]

Verification status

Ref	Type	Metadata verified	DOI/ISBN
[1]	Journal (Neural Networks)	yes	DOI 10.1016/j.neunet.2012.02.016
[2]	Conference (IJCNN)	yes	—
[3]	Conference (CVPR)	yes	DOI 10.1109/CVPR.2016.91
[4]	Conference (NeurIPS/NIPS)	yes	—
[5]	Conference (ECCV/LNCS)	yes	DOI 10.1007/978-3-319-46448-0_2
[6]	Magazine (Sky & Telescope)	yes (page 158–159)	—
[7]	Conference (KDD)	yes	—
[8]–[13]	Official documentation	canonical sites	—
[14]	Official report (WHO)	yes	ISBN 978-92-4-008651-7
[15]	Technical standard (gps.gov)	yes (5th ed., Apr 2020)	—
[16]	Journal (FGCS)	yes	DOI 10.1016/j.future.2022.05.014

Notes: [2], [4], [7] are cited without a DOI (none confirmed during preparation). [6]'s

exact page (158 vs. 159) should be finalised against the print/ADS record. [15] is the

5th edition (April 2020), the edition in effect at preparation time; cite the specific

PDF URL if the style requires it.